STA 325 Case Study Report

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Introduction

This case study explores how three factors affect the clustering of inertial particles in turbulent flows such as clouds. In each of 89 simulation runs, our collaborators simulated particles moving through a turbulent flow, identified clusters of particles, and summarized the distribution of cluster volumes at one (Re, Fr, St) setting. The three parameters take the following values:

- **Reynolds number** Re: 90, 224, 398. This measures how strong the turbulence is.
- **Froude number** Fr: 0.052, 0.3, ∞. This measures the strength of gravity. Smaller Fr means stronger gravity, and $Fr = \infty$ means no gravity.
- **Stokes number** St: 14 values between 0.05 and 3. This measures particle inertia. Particles with small St follow the flow closely, while particles with large St are heavier and respond more slowly.

Our collaborators have two goals:

1. **Prediction:** predict the mean, standard deviation, skewness, and kurtosis of the cluster-volume distribution at new (Re, Fr, St) settings. A single simulation can take days, while a fitted model returns a prediction in seconds, so a good model lets our collaborators explore new settings without running new simulations.
2. **Inference:** determine how turbulence (Re), gravity (Fr), and particle inertia (St) affect the size and shape of the cluster-volume distribution, including whether each effect is nonlinear and whether the effect of one parameter depends on another, and explain these effects physically.

Our final model is as follows, where Re_{224}, Re_{398}, $Fr_{0.3}$, and Fr_∞ are 0/1 indicators for each level (baseline $Re = 90$, $Fr = 0.052$).

$$\begin{aligned}\log \text{responses} = \beta_0 + \beta_1 \log St + \beta_2 (\log St)^2 + \beta_3 (\log St)^3 + \beta_4 Re_{224} + \beta_5 Re_{398} + \beta_6 Fr_{0.3} + \beta_7 Fr_\infty \\ + \beta_8 Re_{224} Fr_{0.3} + \beta_9 Re_{224} Fr_\infty + \beta_{10} Re_{398} Fr_\infty + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2)\end{aligned}$$

For prediction, the model achieves a LOOCV MSE of 0.088 averaged across the 4 responses. For inference, the $Re \times Fr$ terms describe how the effect of gravity changes with turbulence, and the polynomial terms capture the nonlinear effect of St.

Table 1: LOOCV MSE (log scale) for each candidate model. For the GAM, the parameter count is its effective degrees of freedom for the mean.

Model	Parameters	Mean	Standard deviation	Skewness	Kurtosis
Baseline (additive)	6	0.0192	0.7338	0.3561	1.3252
Interaction Model 1	13	0.0104	0.3290	0.0715	0.2119
Interaction Model 2	9	0.0114	0.2565	0.0554	0.1656
Cubic Model	11	0.0093	0.1377	**0.0480**	**0.1567**
GAM (tuned)	12.2	**0.0091**	**0.1367**	0.0495	0.1591

Methodology

Responses and transformations

Each run reports the first four raw moments of its cluster-volume distribution, which we converted to central moments with `raw2central` and then to the mean, standard deviation, skewness, and kurtosis. The four responses differ in skewness: the mean and standard deviation are strongly right-skewed, while skewness and kurtosis are only mildly skewed, so a single log transform might not suit all four. We therefore fit a separate Box-Cox transformation to each response. Every 95% interval for λ contains 0, the log transform (Table 2), so the data support logging all four responses. We also log-transformed St, which is right-skewed and spans a 60-fold range. Re and Fr each take only 3 values, so we treat them as categorical.

Model selection

We compared four linear models of increasing flexibility using leave-one-out cross-validation (LOOCV), which refits each model 89 times with one run held out and measures the error in predicting that run (Table 1). The baseline model has additive effects of $\log St$, Re, and Fr. Interaction Model 2 adds $Re \times Fr$, so each combination has its own level. Interaction Model 1 also lets the slope in $\log St$ differ by Re and Fr. The cubic model adds $(\log St)^2$ and $(\log St)^3$ to Interaction Model 2, so the effect of St can curve. Adding $Re \times Fr$ gave the largest improvement, cutting average LOOCV MSE from 0.609 to 0.122. Adding curvature in St lowered it further to 0.088, while letting the St slope vary by Re and Fr did not help on average (0.156). The cubic model had the lowest LOOCV MSE of the linear models for all four responses.

To check whether the polynomial distorts the effect of St, we also fit a generalized additive model (GAM) that replaces the polynomial with a smoothing spline in $\log St$. Like the cubic model, it uses one curve shared across (Re, Fr) combinations plus a level for each combination; separate curves for each combination would overfit, since each has only 9 to 14 runs. We chose its smoothness by generalized cross-validation. The tuned GAM used 11 to 13 effective degrees of freedom and predicted about as well as the cubic model (average LOOCV MSE of 0.089 versus 0.088), with nearly the same St curve (Figure 2). We kept the cubic model, whose coefficients are simpler to report.

Diagnostics

The cubic model fits all 4 responses well, with R^2 between 0.974 and 0.999. In the baseline model's residuals, whole (Re, Fr) groups sit above or below 0 for skewness and kurtosis, which shows that the effect of Re depends on Fr; the cubic model removes these patterns (Figure 3 in the Appendix), although some uneven spread remains for standard deviation. The largest negative residuals mostly come from runs at the smallest Stokes number, $St = 0.05$, where the model predicts values that are too high. Runs with the same Re and Fr likely come from the same simulated flow, so they may not be fully independent, which could make our intervals slightly too narrow.

Results

Predictions on the hold-out set

We predicted the four statistics at the 23 hold-out settings, with 95% prediction intervals computed on the log scale and then exponentiated, so point predictions estimate medians (all values are in `predictions.csv`). Every hold-out setting lies within the training range, so no prediction requires extrapolation, although the two settings at $St = 0.05$ sit at the edge of the design. Precision varies by statistic: the upper bounds are typically about 1.4 times the lower bounds for the mean, 2.2 times for skewness, 3.9 times for standard deviation, and 4.1 times for kurtosis.

Effects of each parameter

Because the model is fit on the log scale, exponentiated contrasts are multiplicative changes, and because Re and Fr interact, the effect of gravity depends on Re. Figure 1 shows the fitted value of each statistic for every (Re, Fr) combination.

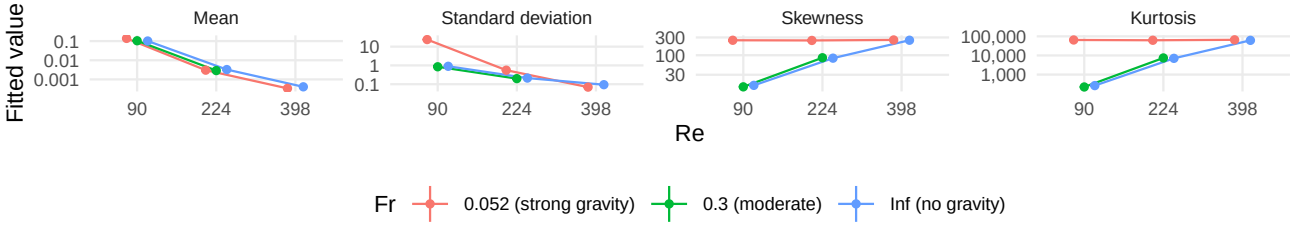


Figure 1: Fitted values at St = 1 for each (Re, Fr) combination, with 95% confidence intervals (log scale). Lines connect the same gravity level across turbulence levels; there are no runs at Re = 398, Fr = 0.3.

Turbulence intensity (Re) controls cluster size. Under strong gravity, the cubic model estimates that increasing Re from 90 to 398 shrinks mean cluster volume by about 410 times and standard deviation by about 340 times. With no gravity, the reductions are about 250 times for the mean and 9.6 times for standard deviation. Stronger turbulence produces a wider range of eddy sizes, including much smaller ones, which appears to break particles into much smaller clusters.

Strong gravity creates heavy tails, but only when turbulence is weak. At $Re = 90$, moving from strong gravity ($Fr = 0.052$) to no gravity ($Fr = \infty$) reduces skewness by a factor of about 16 and kurtosis by about 244, so strong gravity is associated with many small clusters and occasional very large ones. This effect weakens as Re increases: removing gravity reduces kurtosis by about a

factor of 9 at $Re = 224$ and has little effect at $Re = 398$, visible in Figure 1 as a gap at $Re = 90$ that closes as Re increases. One physical interpretation is that as turbulence strengthens, turbulent velocity fluctuations overwhelm gravitational settling, so the flow rather than gravity determines how particles cluster.

Moderate gravity behaves like no gravity. The fitted values for $Fr = 0.3$ and $Fr = \infty$ are similar across the four responses, so within the range studied, gravity is associated with clustering mainly when it is strong. Because there are no runs at $Fr = 0.3$ with $Re = 398$, this comparison covers only the two lower turbulence levels.

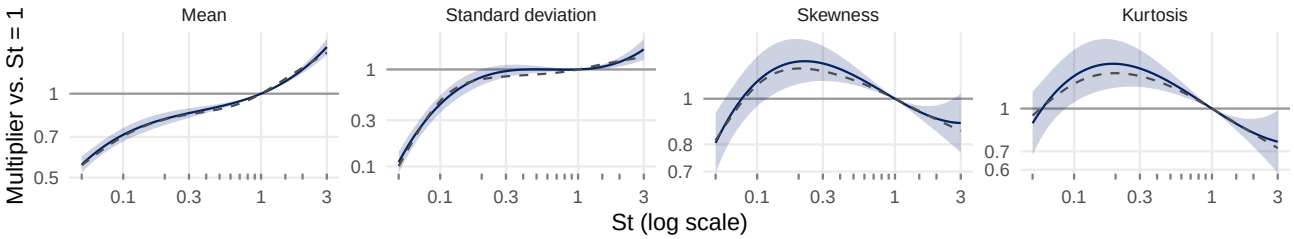


Figure 2: Estimated effect of St on each statistic, as a multiplier relative to St = 1, from the cubic model (solid line, 95% confidence band) and the tuned GAM (dashed line). Tick marks show the observed St values.

Particle inertia (St) has a smaller, nonlinear effect. Figure 2 shows the effect of St relative to $St = 1$. Mean cluster volume rises with St, the standard deviation rises steeply at low St and then levels off, and skewness and kurtosis peak at intermediate St before declining. The cubic and GAM curves agree closely, so these conclusions do not depend on the polynomial form, and the bands widen near $St = 0.05$, where the model fits least well. Because the slope changes across the range, no single "doubling St" percentage summarizes the effect;

Conclusion

Each parameter shapes the cluster-size distribution in a different way. Turbulence intensity is strongly associated with cluster scale, while strong gravity is associated with heavier tails when turbulence is weak. The effect of particle inertia is smaller and changes across the observed St range. Because gravity's influence varies with turbulence strength, conclusions drawn at one turbulence level should not be assumed to hold at another.

Limitations and future work

The model can only predict at the 3 simulated levels of Re and Fr: because it treats them as categorical, it cannot interpolate between levels or extrapolate beyond them, and snapping a new input to the nearest level would ignore the difference between levels. Future simulations should therefore be spread off the grid. The current grid spends all 89 runs on 3 levels each of Re and Fr, and refining it scales badly: adding a fourth level of each parameter raises a full grid from $3^3 = 27$ to $4^3 = 64$ settings. A space-filling design gives every run new values of all three parameters, so the same budget covers far more of the input space and lets Re and Fr be treated as continuous.

Appendix

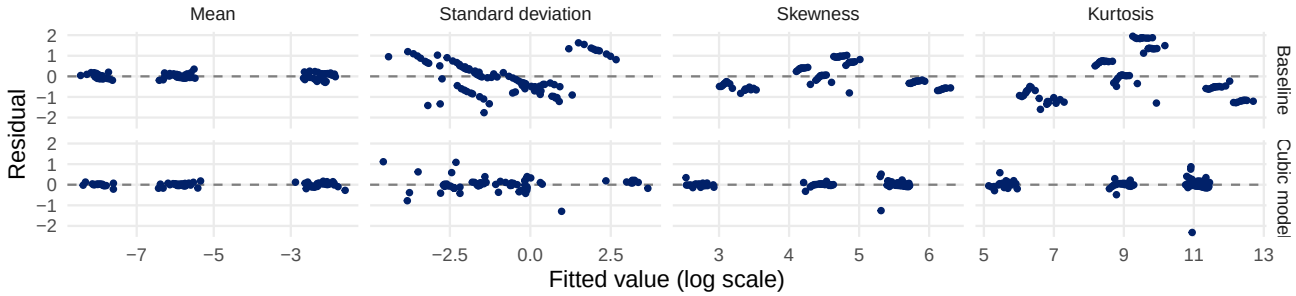


Figure 3: Residuals against fitted values for the baseline model (top) and the cubic model (bottom).

Table 2: Estimated Box-Cox λ for each response, with 95% profile-likelihood intervals. $\lambda = 0$ is the log transform.

Response	Estimate	95% interval
Mean	-0.02	[-0.05, 0.01]
Standard deviation	-0.03	[-0.07, 0.01]
Skewness	-0.11	[-0.28, 0.07]
Kurtosis	-0.05	[-0.12, 0.02]